

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 33

Title: Water soluble vitamins

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Required Prereading and Objectives

- Before DM Lecture 33

Objectives: DLA: Fat soluble vitamins (A, D, E and K)

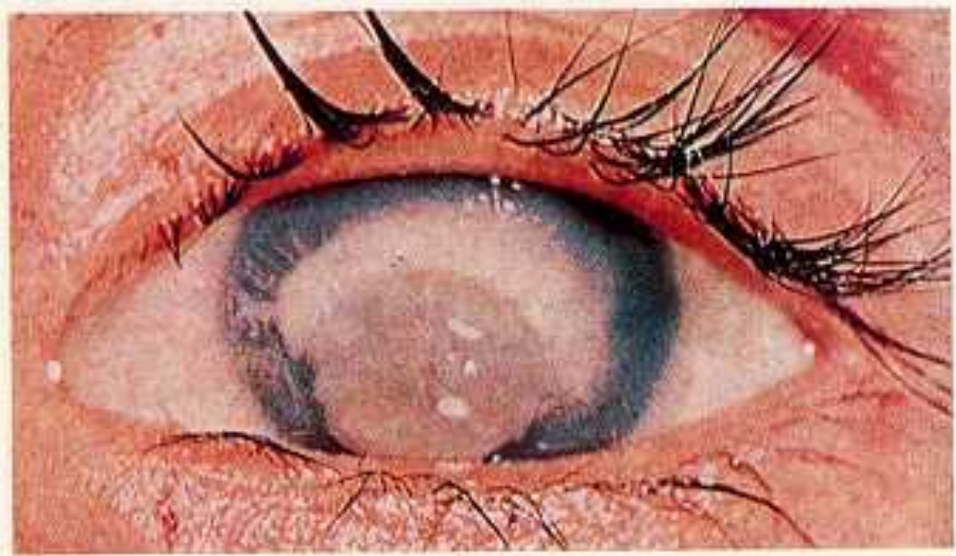
SOM. MK.III.BPM2.3.DM.3.BCHM.1302	Predict the reason for occurrence of fat-soluble vitamin deficiency in patients with fat malabsorption (cystic fibrosis, obstructive jaundice)
SOM. MK.III.BPM2.3.DM.3.BCHM.1303	Review the role of vitamin K in the maturation of clotting factors in the liver.
SOM. MK.III.BPM2.3.DM.3.BCHM.1304	Predict the risk factors (newborn, antibiotics), causes and effects of vitamin K deficiency.
SOM. MK.III.BPM2.3.DM.3.BCHM.1305	Discuss mechanism of inhibition of gamma-carboxylation by warfarin.
SOM. MK.III.BPM2.3.DM.3.BCHM.1306	Describe the functions and effects of dietary vitamin A deficiency. Discuss use of retinoids as drugs in medicine
SOM. MK.III.BPM2.3.DM.3.BCHM.1307	Explain why retinoic acid cannot be used to treat night blindness.
SOM. MK.III.BPM2.3.DM.3.BCHM.1308	Describe formation of vitamin D in skin and the formation of calcitriol (role of liver and kidney)
SOM. MK.III.BPM2.3.DM.3.BCHM.1309	Describe functions of 1,25 dihydroxy-D (calcitriol) related to calcium and phosphate metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1310	Indicate vitamins that act as steroid hormones (calcitriol and retinol)
SOM. MK.III.BPM2.3.DM.3.BCHM.1311	Indicate clinical features and manifestations of hypervitaminosis A and hypervitaminosis D.
SOM. MK.III.BPM2.3.DM.3.BCHM.1312	Describe risk factors and molecular basis of clinical features in rickets and osteomalacia.

Objectives: Water soluble Vitamins

SOM. MK.III.BPM2.3.DM.3.BCHM.1313	Compare and contrast fat-soluble and water-soluble vitamins regarding storage, absorption and risk factors for deficiency
SOM. MK.III.BPM2.3.DM.3.BCHM.1314	Discuss the metabolic basis, clinical manifestations and laboratory tests for thiamine deficiency (B1) in beri-beri and Wernicke Korsakoff syndrome.
SOM. MK.III.BPM2.3.DM.3.BCHM.1315	Indicate coenzyme forms, functions and clinical manifestations of riboflavin deficiency
SOM. MK.III.BPM2.3.DM.3.BCHM.1316	Recapitulate risk factors and metabolic basis of clinical manifestations in niacin deficiency (pellagra).
SOM. MK.III.BPM2.3.DM.3.BCHM.1317	Correlate metabolic basis for clinical manifestations of pyridoxine deficiency (B6). Review significance of B6 supplementation in patients on INH (isoniazid) therapy.
SOM. MK.III.BPM2.3.DM.3.BCHM.1318	Correlate metabolic basis for clinical features of vitamin C deficiency.
SOM. MK.III.BPM2.3.DM.3.BCHM.1319	Describe the utility of vitamin supplementation in management of patients with inherited metabolic disorders. ***

Recommended Reading

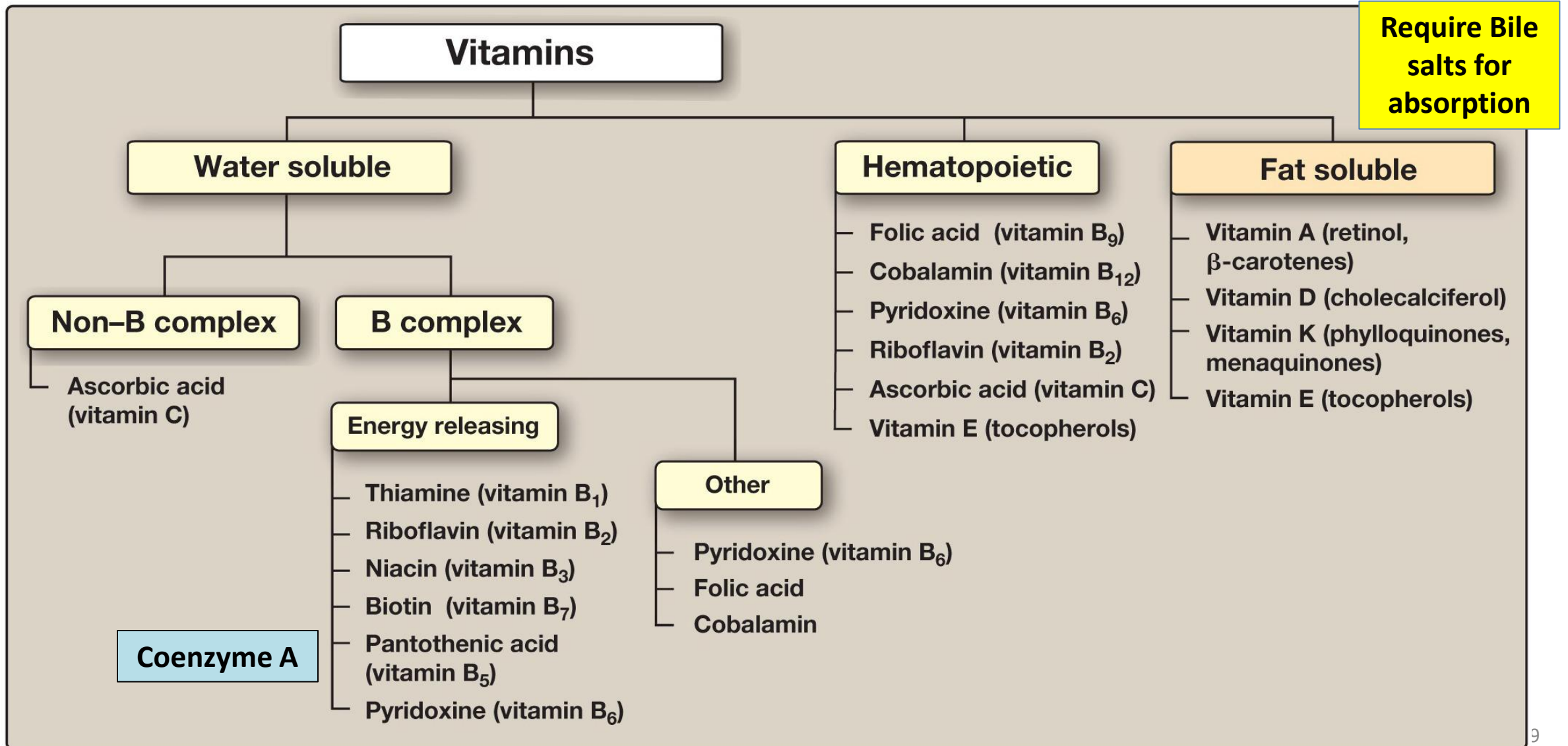
- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Lippincott's Biochemistry; Chapter 28 – Micronutrients: Vitamins
[Micronutrients: Vitamins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Lippincott's Biochemistry Fig 28.29 Pg: 442-443 (Vitamins reference)
- Lippincott's Study Q: 28.1 – 28.8
- Lippincott's Study Q Chapter 4: 4.4, 4.5 [Fibrous Proteins | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library \(lwwhealthlibrary.com\)](#)
- **Directed Learning activity – Vitamins A, D, E and K**



VITAMINS

- Organic compounds that cannot be synthesized by humans and must be supplied by diet
- Required for functions like co-enzymes or as hormones
- **Thirteen vitamins**
- Classified based on solubility

Classification of vitamins



Fat soluble Vitamins (Directed Learning activity)

- Absorption depends on normal fat digestion and absorption
- **Maldigestion** and **malabsorption** of dietary fats results in secondary deficiency (bile duct obstruction, cystic fibrosis).
- **Stored in liver** (Vitamin K stored in least amounts)

Vitamin A : Vision, epithelial tissue, growth

Deficiency: Night blindness; bitot spots; Keratomalacia

Vitamin D : Bone mineralization, blood calcium regulation

Deficiency: Rickets in children; Osteomalacia in adults

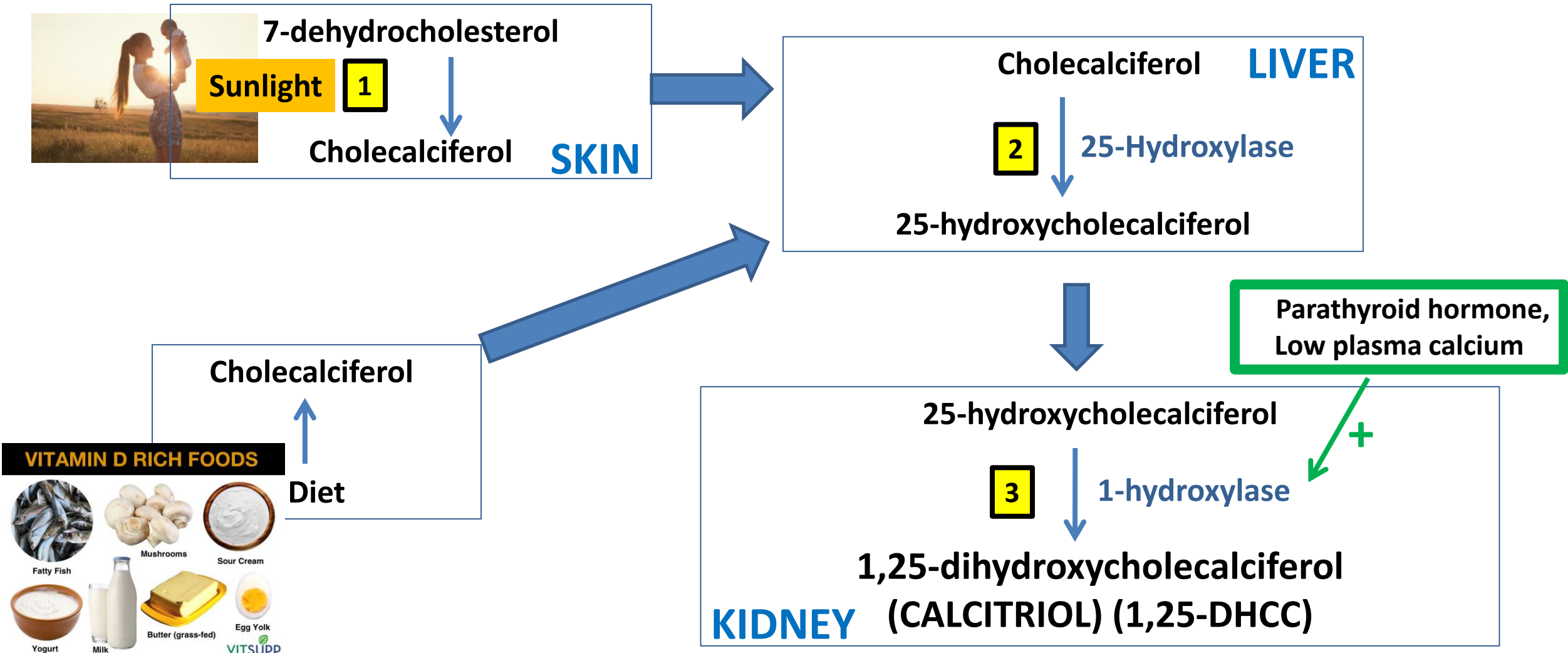
Vitamin E : Lipid soluble anti-oxidant

Deficiency: Hemolysis

Vitamin K : Clotting factor synthesis; Coenzyme for gamma-carboxylase

Deficiency: Prolonged prothrombin time; Increased bruising

Formation of calcitriol (1,25 dihydroxy cholecalciferol) (active form of vitamin D)



Vit D deficiency: Radiologic and lab findings



Lab findings

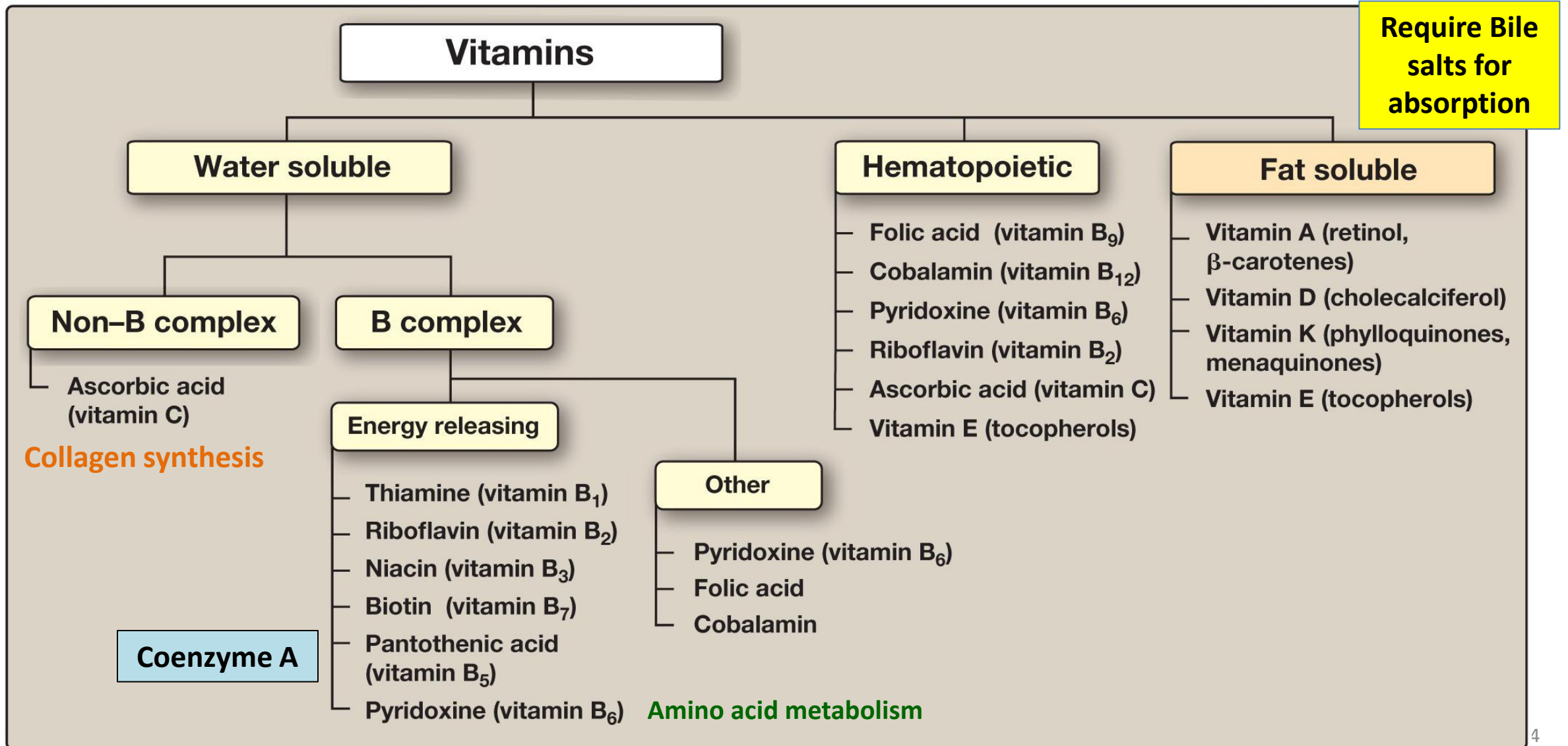
- Elevated **Serum alkaline phosphatase (ALP)**
- **Low serum calcium levels**
- **Low serum phosphate levels**

(A) **Rickets.** Anteroposterior radiograph of the wrist and hand in a 3-year-old child with nutritional rickets. The child had been put on a strict diet without dairy products. Note the widening, cupping, and fraying of the distal radius (arrowhead) and ulna metaphyses with an associated increase in the thickness of the growth plate (arrow). These changes are the consequence of disordered endochondral growth.

(B) **Normal.** Radiograph of the hand of a healthy 3-year-old child, without rickets.

Water soluble Vitamins

Usually NOT stored – daily supplements essential (Except vitamin B12 stored in liver)



Vitamin	Other names	Active form	Function	Deficiency	Signs and symptoms	Toxicity	Notes
Vitamin B ₉	Folic acid	Tetrahydro-folic acid	Transfer one-carbon units; synthesis of methionine, serine, purine nucleotides, and thymidine monophosphate	Megaloblastic anemia Neural tube defects	Anemia Birth defects	None	Administration of high levels of folate can mask vitamin B ₁₂ deficiency
Vitamin B ₁₂	Cobalamin	Methylcobalamin Deoxyadenosyl cobalamin	Coenzyme for reactions: Homocysteine → methionine Methylmalonyl CoA → succinyl CoA	Pernicious anemia Dementia Spinal degeneration	Megaloblastic anemia Neuropsychiatric symptoms	None	Pernicious anemia is treated with intramuscular or high-dose oral vitamin B ₁₂
Vitamin C	Ascorbic acid	Ascorbic acid	Antioxidant Dopamine → norepinephrine Coenzyme for hydroxylation reactions, for example: In procollagen: Proline → hydroxyproline Lysine → hydroxylysine	Scurvy	Sore, spongy gums Loose teeth Poor wound healing Bleeding	None	Benefits of supplementation not established in controlled trials
Vitamin B ₆	Pyridoxine Pyridoxamine Pyridoxal	Pyridoxal phosphate	Coenzyme for enzymes, particularly in amino acid metabolism Heme synthesis	Rare	Glossitis Neuropathy	Yes	Deficiency can be induced by isoniazid Sensory neuropathy occurs at high doses
Vitamin B ₁	Thiamine	Thiamine pyrophosphate	Coenzyme of enzymes catalyzing: Pyruvate → acetyl CoA α-Ketoglutarate → Succinyl CoA Ribose 5-P + xylulose 5-P → Sedoheptulose 7-P + Glyceraldehyde 3-P Branched-chain α-keto acid oxidation	Beriberi Wernicke-Korsakoff syndrome (most common in alcohol use disorder)	Peripheral neuropathy (dry form), edema and cardiomyopathy (wet form) Confusion, ataxia, memory loss, hallucinations, dysregulated eye movements	None	—
Vitamin B ₃	Niacin Nicotinic acid	NAD ⁺ , NADP ⁺	Electron transfer	Pellagra	Dermatitis Diarrhea Dementia	None	High doses of niacin used to treat hyperlipidemia
Vitamin B ₂	Riboflavin	FMN, FAD	Electron transfer	Rare	Dermatitis Angular stomatitis	None	—
Vitamin B ₇	Biotin	Enzyme-bound biotin	Carboxylation reactions Pyruvate → oxaloacetate Acetyl CoA → malonyl CoA Propionyl Co → methylmalonyl CoA 3-Methylcrotonyl CoA → 3-methylglutaconyl CoA	Rare	Dermatitis	None	Consumption of large amounts of raw egg whites (which contains a protein, avidin, that binds biotin) can induce a biotin deficiency
Vitamin B ₅	Pantothenic acid	Coenzyme A	Acyl carrier	Rare	—	None	—
WATER SOLUBLE				FAT SOLUBLE			
Vitamin A	Retinol Retinal Retinoic acid β-Carotene	Retinol Retinal Retinoic acid	Maintenance of reproduction Vision Promotion of growth Differentiation and maintenance of epithelial tissues Gene expression	Night blindness Xerophthalmia Infertility Growth retardation	Increased visual threshold Dryness of cornea	Yes	β-Carotene not acutely toxic, but supplementation is not recommended Excess vitamin A can increase incidence of fractures
Vitamin D	Cholecalciferol Ergocalciferol	1,25-Dihydroxy-cholecalciferol	Calcium uptake Gene expression	Rickets (in children) Osteomalacia (in adults)	Soft, pliable bones	Yes	Vitamin D is not a true vitamin because it can be synthesized in skin; application of sunscreen lotions or presence of dark skin color decreases this synthesis.
Vitamin K	Menadione Menaquinone Phylloquinone	Menadione Menaquinone Phylloquinone	γ-Carboxylation of glutamate residues in clotting and other proteins	Newborn Rare in adults	Bleeding	Rare	Vitamin K produced by intestinal bacteria. Vitamin K deficiency common in newborns Intramuscular treatment with vitamin K is recommended at birth
Vitamin E	α-Tocopherol	Any of several tocopherol derivatives	Antioxidant	Rare	Red blood cell fragility leads to hemolytic anemia	None	Benefits of supplementation for disease prevention not established in controlled trials

Vitamin deficiency

Primary deficiency

- Dietary deficiency
- Starvation
- Malnutrition

Secondary deficiency

- Reduced intake
 - Dental problems, chronic disease, morning sickness
- Malabsorption
 - Diarrhea, genetic defects
 - **Post-bariatric surgery**
- Increased requirements
 - Pregnancy, post-operative, periods of rapid growth
- Increased loss
 - Lactation

Water soluble vitamins

- Daily supplements (not stored, except vitamin B12)
- Do **not require bile salts and chylomicrons** for absorption
(Absorption easier than fat soluble vitamins)
- Toxicity not common (excess excreted in urine)

VITAMIN C (Ascorbic acid)

- Water soluble vitamin
- Connective tissue (**collagen synthesis**) and **wound healing**
 - Prolyl and lysyl hydroxylase coenzyme – Hydrogen bond formation → Collagen stability
 - Lysyl oxidase requires copper
- **Absorption of iron:** Ferrous state
 - Low dietary vitamin C → **Microcytic anemia** (reduced iron absorption)
- Water soluble anti-oxidant: Free radical scavenger
 - Antioxidant vitamins C, E and beta-carotene (provitamin A)
 - Reduce chronic disease

Vitamin C Deficiency: Scurvy

- Perifollicular hemorrhages (Fragile blood vessels)
- Sore, spongy gums: Bleeding gums
- Loose teeth
- Bleeding into joints
- Frequent bruising
- Impaired/ delayed wound healing

Connective tissue defects
(**decreased hydroxyl groups**): fewer H-bonds in collagen



Similar to vitamin K deficiency – Molecular mechanism for increased bleeding tendency is very different!!!!



Perifollicular hemorrhages

Earliest sign



L p.377

Figure 6.3 Stomatitis in scurvy, reflecting deficient collagen synthesis (courtesy of Wellcome Medical Photographic Library, no. N0005641C).



Vitamin C deficiency leads to scurvy

VITAMIN B COMPLEX

- Thiamine (B1)
- Riboflavin (B2)
- Niacin (B3)
- Pantothenic acid (Coenzyme A) (B5)
- Pyridoxine (B6)
- Biotin (B7)
- Folic acid (B9)
- Cobalamin (B12)

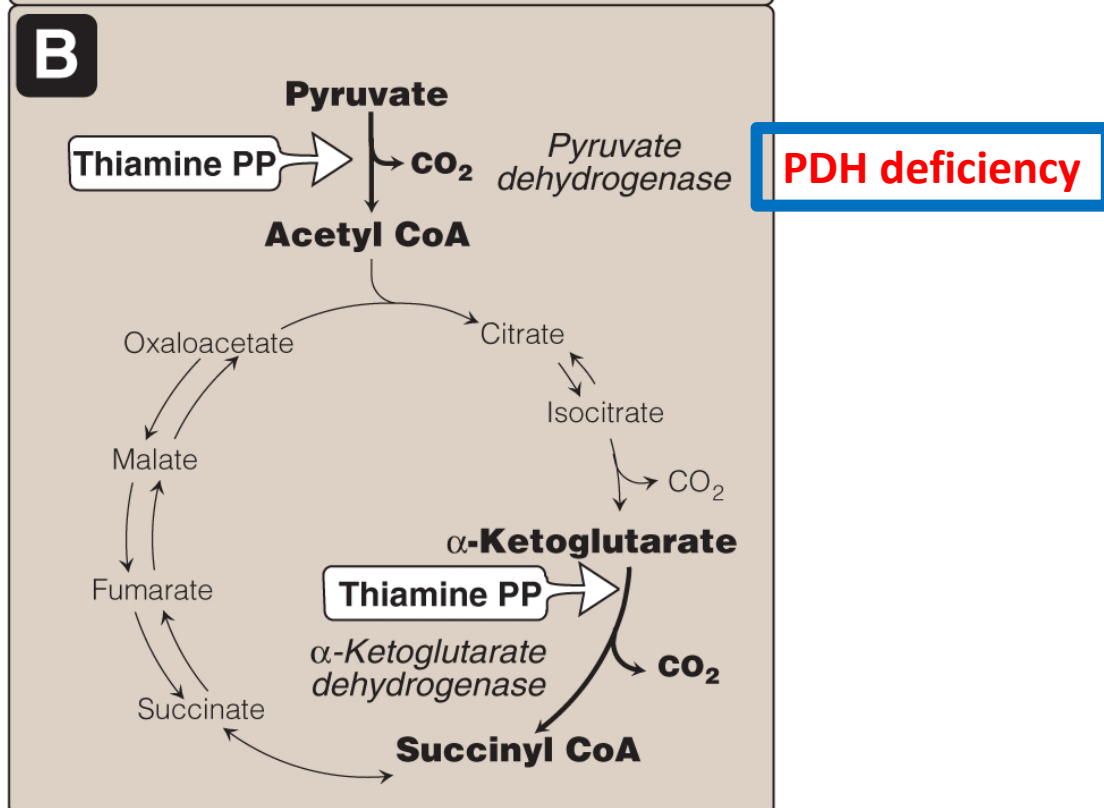
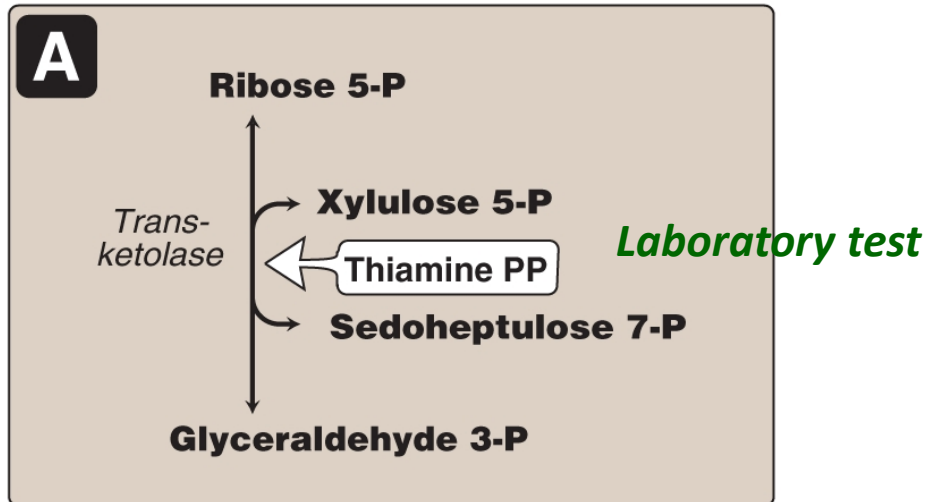
- Identify **coenzymes**
- Identify **reactions** requiring
- Lab tests
- Manifestations of **deficiency**
- Utility in **inherited disorders*****

**Water soluble vitamins NOT stored –
daily supplements essential (Vitamin
B12 stored in liver)**

VITAMIN B1 (THIAMINE)

- Coenzyme: **TPP (Thiamine pyrophosphate)**
- Coenzyme
 - Oxidative decarboxylation of alpha-keto acids – maintains nerve tissue (PDH complex, α -ketoglutarate dehydrogenase, **branched chain α -keto acid dehydrogenase**)
 - Transketolase in pentose phosphate pathway (HMP shunt)
 - B1 supplements: **Maple syrup urine disease (MSUD)*** and PDH deficiency**
- Dietary Sources:
 - Widely available in diet **BUT**
 - Deficient in Refined foods like polished rice, white flour, white sugar

Reactions requiring TPP



Branched chain amino acid



Branched chain keto acid



Branched chain metabolites

MSUD

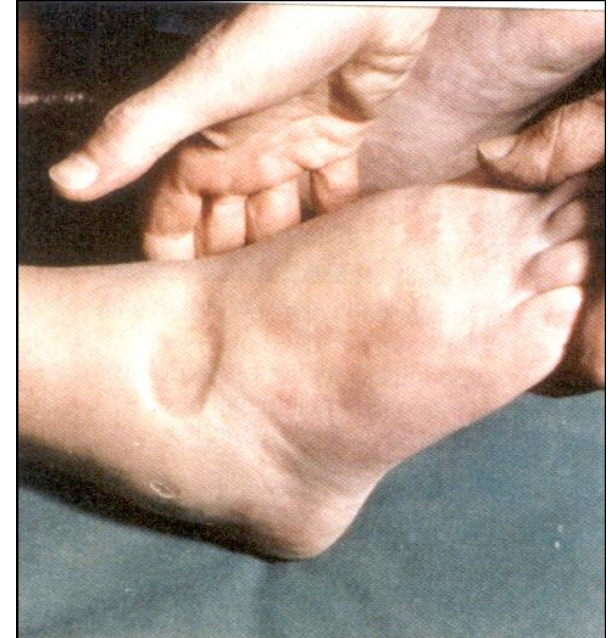
Branched chain keto
acid dehydrogenase

TPP

Thiamine deficiency: **Beriberi**

- Polished rice is staple
- Affects aerobic tissues (Brain and cardiac muscle)
- Polyneuropathy: Disruption of motor, sensory and reflex arcs; progress to paralysis (**Dry beri beri**)
- Cardiovascular: Cardiac failure (**Wet beri beri**)
- **Laboratory tests**
 - *Low erythrocyte transketolase activity*
 - *Serum thiamine levels*

**Pedal edema in
Wet beriberi**



Thiamine deficiency

Wernicke-Korsakoff syndrome:

- Devastating neurologic complication in chronic alcohol use disorder
- Ophthalmoplegia and [nystagmus](#) (to-and-fro eyeball movement)
- Ataxia, confusion, disorientation, and loss of memory
- Confabulation
- Reviewed in Neuro-Behavioral module

VITAMIN B2 (RIBOFLAVIN)

- Coenzyme forms
 - Flavin mononucleotide (FMN)
 - Flavin adenine dinucleotide (FAD)
- Oxidation-reduction reactions of TCA cycle, beta oxidation (Succinate dehydrogenase, PDH, Acyl CoA DH.....)

Signs and symptoms of deficiency:

- Cheilosis: Cracks and fissures at angles of mouth
- Glossitis: Inflammation and atrophy of tongue
- Facial dermatitis

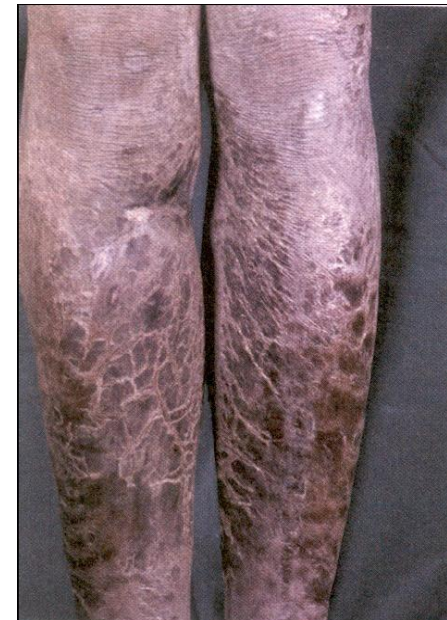


VITAMIN B3 (NIACIN)

- Coenzyme forms
 - NAD^+
 - NADP^+
- **Coenzymes in oxidation-reduction reactions**
 - NAD^+ – Dehydrogenases (PDH, alpha-ketoglutarate dehydrogenase, malate dehydrogenase,)
 - NADP^+ – HMP shunt and fatty acid and cholesterol synthesis (G6PD, HMG CoA reductase, ...)
- **Therapeutic uses:**
 - Inhibits adipose tissue lipolysis and reduces free fatty acids: Type IIb hyperlipoproteinemia

Niacin Deficiency

- **Pellagra** (rough skin): **3Ds**
- **Dermatitis** (skin)
 - Exposed areas of body
 - Redness, thickening and roughening of skin
- **Diarrhea** (GIT)
- **Dementia** (CNS)
 - Neuron degeneration in brain and spinal tracts
 - Loss of memory
- **Death**



Pellagra dermatitis 'necklace-like' and hands and legs

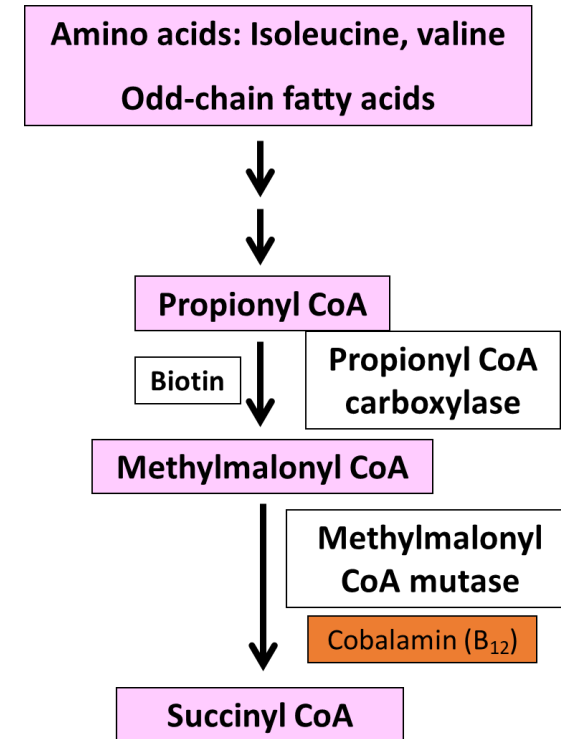


Tryptophan: Precursor of NAD⁺

- **Tryptophan** forms NAD⁺ and NADP⁺ (refer amino acid metabolism)
- Risk factors for Niacin deficiency (Pellagra)
 - **Corn based diets** (Niacin and tryptophan deficient)
 - **Hartnup disease** (Neutral amino acid transporter defect; Less dietary tryptophan absorption and loss of Trp in urine)*** - **Dietary niacin supplements**
 - **Carcinoid syndrome** (excessive tryptophan to serotonin conversion; less tryptophan for NAD⁺ formation)

BIOTIN (VITAMIN B7)

- Carboxylation reactions
- **Multiple carboxylase deficiency*****: Defect incorporating biotin
- Coenzyme for: ABC enzymes (ATP, Biotin, CO₂)
 - Pyruvate carboxylase (gluconeogenesis) → Impaired gluconeogenesis
 - Acetyl-CoA carboxylase (fatty acid synthesis) → Hair loss and dermatitis
 - Propionyl-CoA carboxylase (Conversion of propionyl CoA to methylmalonyl CoA) → **High serum propionate levels**
- **Avidin** (raw egg white), inhibits biotin absorption



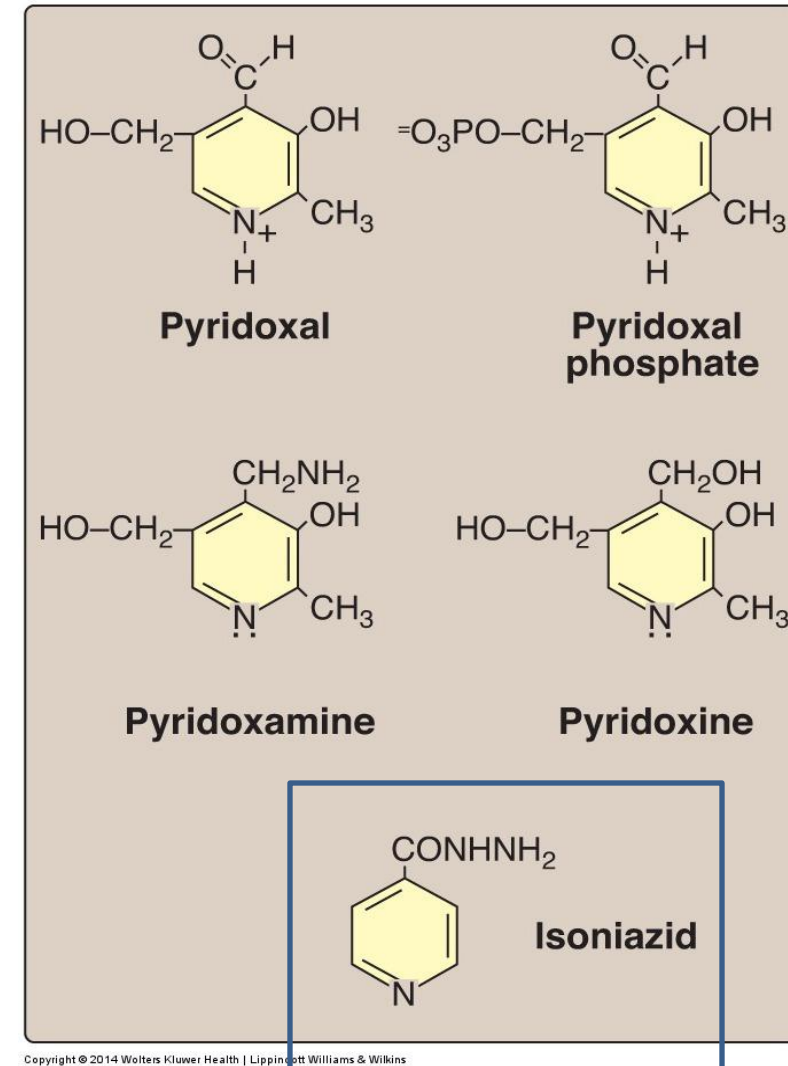
VITAMIN B6 (PYRIDOXINE)

Transamination of amino acids	Aminotransferase (AST and ALT)	Oxaloacetate + glutamate $\leftrightarrow$ Aspartate + alpha ketoglutarate Pyruvate + glutamate $\leftrightarrow$ Alanine+ alpha ketoglutarate
Amino acid decarboxylation (Neurotransmitter synthesis)	Amino acid decarboxylases	Histidine $\rightarrow$ Histamine Glutamate $\rightarrow$ GABA 5-hydroxytryptophan $\rightarrow$ 5-hydroxytryptamine (Serotonin) DOPA $\rightarrow$ Dopamine
Condensation	ALA synthase (Heme synthesis)	Glycine + Succinyl CoA $\rightarrow$ Delta amino levulinic acid
Homocysteine metabolism	Cystathionine synthase	Homocysteine + Serine $\rightarrow$ Cystathionine

Inherited homocystinuria: B6 supplements***

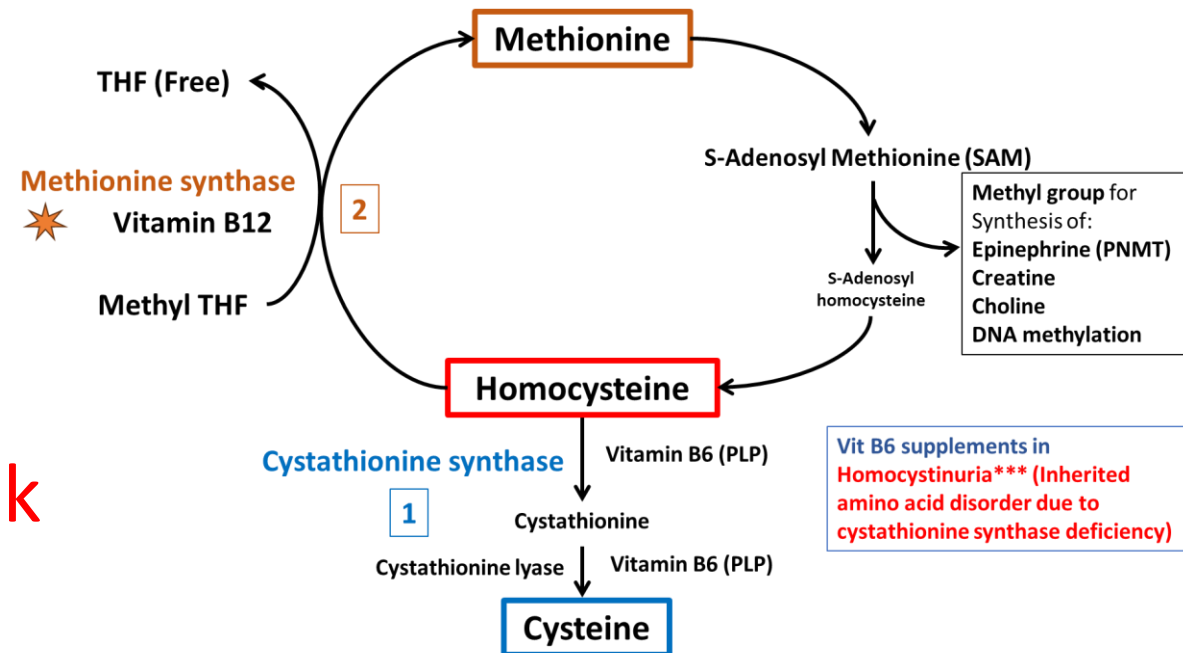
Isoniazid therapy: Risk factor for Pyridoxine Deficiency

- Isoniazid (anti-tuberculosis drug)
 - Inactivates pyridoxine
 - Pyridoxine supplements given with INH
 - Adverse effects if B6 supplements not administered:
 - Peripheral neuropathy (reduced neurotransmitter synthesis) and
 - Microcytic anemia (reduced heme synthesis)



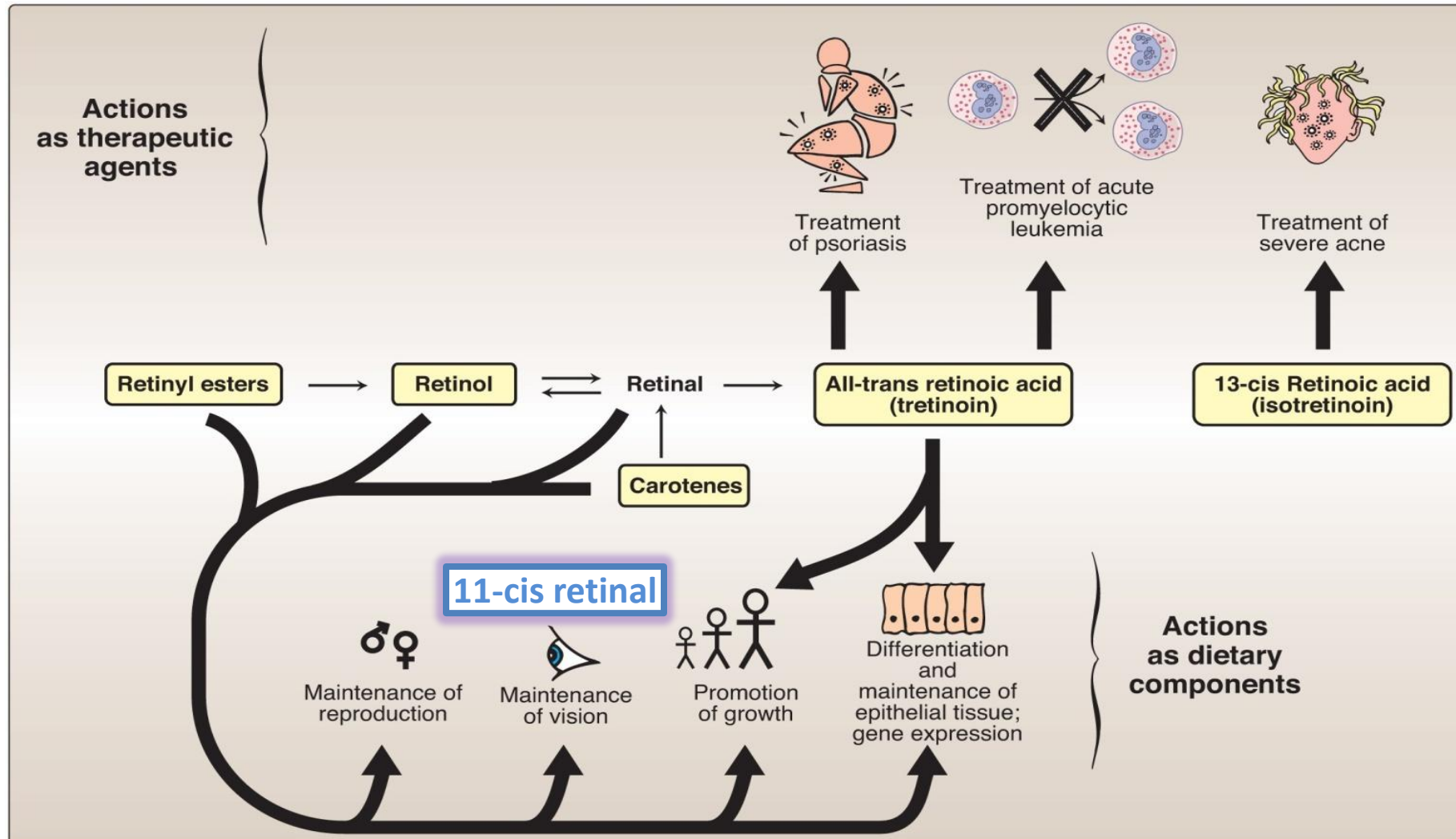
Pyridoxine deficiency: Manifestations

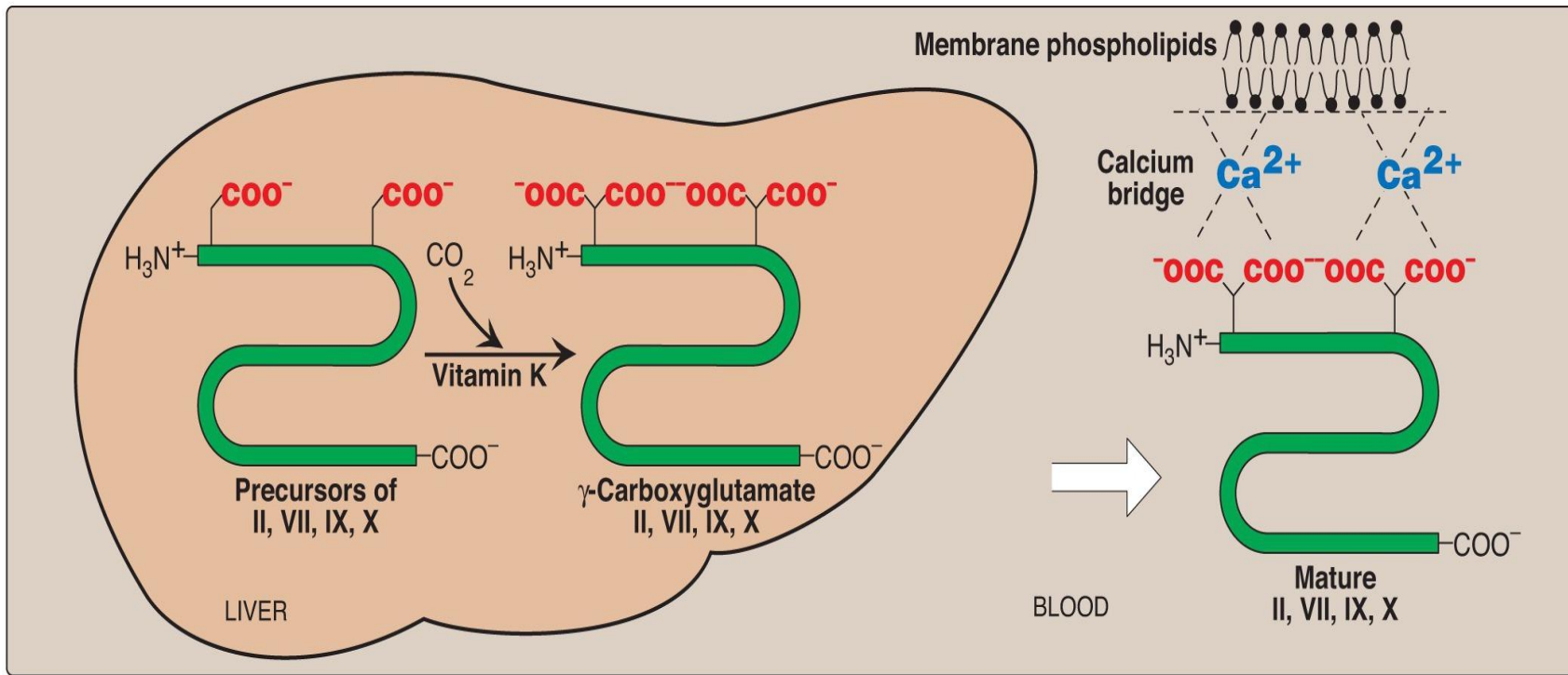
- **Microcytic anemia** (Reduced heme synthesis due to low ALA synthase activity)
- **Peripheral neuropathy** (Reduced neurotransmitter formation)
- Increased **cardiovascular disease risk** (high plasma homocysteine levels)
- **Seizures** in infants (Reduced neurotransmitter formation)



Deficiencies of dietary retinoids leads to night blindness, less fertility, slowed growth and skin problems.

Retinoic acid is used to treat severe acne and psoriasis.





Role of vitamin K

- γ -carboxylation (post-translational modification) forms mature clotting factors that contain γ -carboxyglutamate (Gla) and is capable of subsequent activation
- γ -carboxylation of the clotting factors facilitates **Ca²⁺-binding** because of two adjacent negatively charged carboxylate groups
- The clotting factor calcium complex can then bind to phospholipids on the platelet membrane. (ref: Blood coagulation and fibrinolysis)